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Batch: July 2026

Exploratory Data Analysis of COVID-19 Patients Dataset

Abstract :

This project presents an Exploratory Data Analysis (EDA) of the global COVID-19 pandemic dataset compiled by Johns Hopkins University. Spanning January 22, 2020, to July 27, 2020, the dataset comprises 147,580 daily records across 188 countries and regions, capturing geographical coordinates, case classifications, and daily case tallies. Implemented in Python utilizing Pandas, NumPy, Matplotlib, and Seaborn, the workflow encompasses data cleaning, geospatial correlation analysis, statistical aggregation, and trend visualization. Missing entries in the regional parameters (104,528 null values) were standardized during preprocessing. Key findings indicate a continuous escalation in daily infections, peaking at nearly 140,000 cases in a single day between June and July 2020. Categorical breakdown reveals that total reported infections consist of 61.7% confirmed cases (~16M), 35.8% recoveries (~10M), and a 2.4% mortality rate (<1M). Geospatial assessments demonstrate heavy clustering in latitudes -10 degree to -20 degree (torrid zone) and longitudes -50 degree to -100 degree (Americas). The study provides critical empirical insights into early pandemic progression patterns.

Introduction :

The rapid emergence and widespread transmission of the novel coronavirus (COVID-19) in early 2020 generated vast volumes of epidemiological data worldwide. Understanding the distribution, spatial clustering, and progression of the disease is vital for healthcare institutions, policymakers, and researchers.

Exploratory Data Analysis (EDA) is an essential statistical and visual framework used to uncover underlying patterns, spot anomalies, and test hypotheses with graphical representations. Performing EDA on global pandemic data helps quantify infection growth

rates, evaluate fatality and recovery benchmarks, and determine the geographic dimensions of disease spread across different hemispheres.

Using Python's data analysis and visualization ecosystem (Pandas, NumPy, Matplotlib, and Seaborn), this project models raw records into structured statistical figures, correlation matrices, and time-series distributions.

Problem Statement :

The main objective of this Exploratory Data Analysis is to investigate the global date-wise COVID-19 dataset provided by Johns Hopkins University and extract key public health insights.

The dataset contains daily historical entries from January 22, 2020, to July 27, 2020, covering 188 countries and regions. The key objectives addressed in this study include:

- Determining the monthly infection growth trends globally.
- Evaluating whether geographical parameters (latitude and longitude) correlate with disease concentration.
- Finding the distribution across confirmed cases, recoveries, and mortality rates.
- Measuring inter-feature dependencies through correlation heatmaps and pair plots.

Results and Discussion :

The exploratory analysis revealed distinct seasonal and geographic patterns across the 147,580 records:

- **Geographical Distribution:** Regions located between latitudes -10 degree and -20 degree (torrid zone) exhibited elevated case concentrations. Longitudinal analysis showed maximum clustering between -50 degree and -100 degree, corresponding heavily to outbreaks in the United States and the Americas.
- **Timeline Surge:** Monthly bar chart analysis showed sustained growth from January through July 2020, with daily global tallies peaking near 140,000 cases in a single day between June and July.
- **Case Proportion:** Across all logged entries, confirmed cases formed **61.7%** (~16M), recovered patients constituted **35.8%** (~10M), and fatalities remained constrained at **2.4%** (<1M).

Conclusion :

In this project, an in-depth exploratory analysis was conducted on the global COVID-19 pandemic dataset. Data cleaning successfully handled over 104,000 null regional entries, enabling robust analysis across all 188 countries. The visualizations and statistical aggregations mapped transmission patterns, confirmed a high recovery rate against a 2.4% mortality rate, and identified key geographic epicenters during the initial six months of the pandemic.

In the future, this analysis can be expanded by integrating predictive machine learning models (such as ARIMA, Prophet, or Random Forest regressors) and deploying an interactive dashboard using Streamlit or Flask for real-time monitoring and resource planning.

References:

1. Johns Hopkins University Center for Systems Science and Engineering (CSSE) COVID-19 Dataset.
2. Rami Krispin, *Coronavirus Dataset Repository*, GitHub, 2020.